

The NASA Innovative Advanced Concepts (NIAC), formerly NASA Institute for Advanced Concepts (NIAC), is a NASA program for development of far reaching, long term advanced concepts by "creating breakthroughs, radically better or entirely new aerospace concepts". It funds work on revolutionary aeronautics and space concepts that can dramatically impact how NASA develops and conducts its missions. The program operated under the name NASA Institute for Advanced Concepts from 1998 until 2007 (managed by the Universities Space Research Association on behalf of NASA), and was reestablished in 2011 under the name NASA Innovative Advanced Concepts and continues to the present.

History

The NASA Institute for Advanced Concepts (NIAC) was a NASA-funded program that was operated by the Universities Space Research Association (USRA) for NASA from 1998 until its closure on 31 August 2007. NIAC was to serve as "an independent open forum, a high-level point of entry to NASA for an external community of innovators, and an external capability for analysis and definition of advanced aeronautics and space concepts to complement the advanced concept activities conducted within NASA."

NIAC sought proposals for revolutionary aeronautics and space concepts to dramatically impact how NASA developed and conducted its missions. NIAC provided a highly visible, recognizable, and high-level entry point for outside thinkers and researchers. It encouraged proposers to think decades into the future in pursuit of concepts that would "leapfrog" the evolution of contemporary aerospace systems. These concepts were expected to be based on sound scientific principles and attainable within a 10 to 40-year time frame.

NIAC received a total of 1,309 proposals and awarded 126 Phase I grants and 42 Phase II contracts for a total value of \$27.3 million from February 1998 to 2007.

NASA announced on March 1, 2011 that the NIAC concept would be re-established at NASA with similar goals, maintaining the acronym NIAC.

NIAC 1998–2007

Studies funded by the original NIAC 1998–2007 include

Bio-Nano-Machines for Space Applications – Constantinos Mavroidis

System Feasibility Demonstrations of Caves and Subsurface Constructs for Mars Habitation and Scientific Exploration (Caves of Mars Project) – Penelope J. Boston

Lunar space elevator – Jerome Pearson – [final report.pdf](#)

Magnetic sail – Robert Zubrin

Mars Entomopter – Robert Michelson / Anthony Colozza – [Phase II final report.pdf](#)

Mini-magnetospheric plasma propulsion – Robert M. Winglee

Momentum exchange tether – Thomas J. Bogar – [final report.pdf](#)

New Worlds Mission – Webster Cash

Space elevator – Bradley C. Edwards

Closing of the original NIAC

On July 2, 2007, NIAC announced that "NASA, faced with the constraints of achieving the Vision for Space Exploration, has made the difficult decision to terminate NIAC, which has been funded by NASA since inception. Effective August 31, 2007, the original NIAC organization ceased operations.

Revised NIAC

Following the termination of the original NIAC program, Congress requested a review of the NIAC program by the United States National Research Council (NRC) of the National Academy of Sciences. The review was done in 2009, and concluded that in order to achieve its mission, NASA needs "a mechanism to investigate visionary, far-reaching advanced concepts," and

recommended that NIAC, or a NIAC-like program, should be reestablished. Consistent with this recommendation, it was announced on March 1, 2011 that the NIAC was to be revived with similar goals leading to the establishment in 2011 of a project within the NASA Office of Chief Technologist, the NASA Innovative Advanced Concepts, maintaining the acronym NIAC. It is now part of the NASA Space Technology Mission Directorate (STMD).

According to Michael Gazarik, director of NASA's Space Technology Program, "Through the NASA Innovative Advanced Concepts program, NASA is taking the long-term view of technological investment and the advancement that is essential for accomplishing our missions. We are inventing the ways in which next-generation aircraft and spacecraft will change the world and inspiring Americans to take bold steps."

2011 NIAC Project Selections

The revived NIAC, with the slightly-changed name "NASA Innovative Advanced Concepts," funded thirty phase-I studies in 2011 to investigate advanced concepts.

Duda, Kevin: Variable Vector Countermeasure Suit (V2Suit) for Space Habitation and Exploration

Ferguson, Scott: Enabling All-Access Mobility for Planetary Exploration Vehicles via Transformative Reconfiguration

Gilland, James: The Potential for Ambient Plasma Wave Propulsion

Gregory, Daniel: Space Debris Elimination (SpaDE)

Hogue, Michael: Regolith Derived Heat Shield for a Planetary Body Entry and Descent System with In-Situ Fabrication

Hohman, Kurt: Atmospheric Breathing Electric Thruster for Planetary Exploration

Howe, Steven: Economical Radioisotope Power

Khoshnevis, Behrokh: Contour Crafting Simulation Plan for Lunar Settlement Infrastructure

Build-Up

Kwiat, Paul: Entanglement-assisted Communication System for NASA's Deep-Space Missions: Feasibility Test and Conceptual Design

Mankins, John: SPS-ALPHA: The First Practical Solar Power Satellite via Arbitrarily Large PHased Array

Miller, David: High-temperature Superconductors as Electromagnetic Deployment and Support Structures in Spacecraft

Paul, Michael: Non-Radioisotope Power Systems For Sunless Solar System Exploration Missions

Pavone, Marco: Spacecraft/Rover Hybrids for the Exploration of Small Solar System Bodies

Ritter, Joe: Ultra-Light "Photonic Muscle" Space Structures

Scott, Gregory: Low Power Microrobotics Utilizing Biologically Inspired Energy Generation

Short, Kendra: Printable Spacecraft

Sibille, Laurent: In-Space Propulsion Engine Architecture based on Sublimation of Planetary Resources: from exploration robots to NEO mitigation

Silvera, Isaac: Metallic Hydrogen: A Game Changing Rocket Propellant

Slough, John: Nuclear Propulsion through Direct Conversion of Fusion Energy

Staehle, Robert: Interplanetary CubeSats: Opening the Solar System to a Broad Community at Lower Cost

Stekalov, Dmitry: Ghost Imaging of Space Objects

Stysley, Paul: Laser-Based Optical Trap for Remote Sampling of Interplanetary and Atmospheric Particulate Matter

Swartzlander, Grover: Steering of Solar Sails Using Optical Lift Force

Tarditi, Alfonso: Aneutronic Fusion Spacecraft Architecture

Thibeault, Sheila: Radiation Shielding Materials Containing Hydrogen, Boron, and Nitrogen: Systematic Computational and Experimental Study

Tripathi, Ram: Meeting the Grand Challenge of Protecting Astronaut's Health: Electrostatic Active Space Radiation Shielding for Deep Space Missions

Werka, Robert: Proposal for a Concept Assessment of a Fission Fragment Rocket Engine (FFRE) Propelled Spacecraft

Westover, Shayne: Radiation Protection and Architecture Utilizing High Temperature Superconducting Magnets

Whittaker, William: Technologies Enabling Exploration of Skylights, Lava Tubes and Caves

Wie, Bong: Optimal Dispersion of Near-Earth Objects

2012 NIAC project selections

In August 2012, NIAC announced selection of 18 new phase-I proposals, along with Phase-II grants for continuation of 10 projects selected in earlier solicitations. These include many projects ranging from Landsailing rovers on Venus to schemes to explore under the ice of Europa. Phase I projects selected were:

Agogino, Adrian: Super Ball Bot - Structures for Planetary Landing and Exploration

Arrieta, Juan: The Regolith Biters: A Divide-And-Conquer Architecture for Sample-Return Missions

Cohen, Marc: Robotic Asteroid Prospector (RAP) Staged from L-1: Start of the Deep Space Economy

Ditto, Thomas: HOMES - Holographic Optical Method for Exoplanet Spectroscopy

Flynn, Michael: Water Walls: Highly Reliable and Massively Redundant Life Support Architecture

Gellett, Wayne: Solid State Air Purification System

Hoyt, Robert: NanoTHOR: Low-Cost Launch of Nanosatellites to Deep Space

Hoyt, Robert: SpiderFab: Process for On-Orbit Construction of Kilometer-Scale Apertures

Kirtley, David: A Plasma Aerocapture and Entry System for Manned Missions and Planetary Deep Space Orbiters

Landis, Geoffrey: Venus Landsailing Rover

Lantoine, Gregory: MAGNETOUR: Surfing Planetary Systems on Electromagnetic and Multi-Body Gravity Fields

McCue, Leigh: Exploration of Under-Ice Regions with Ocean Profiling Agents (EUROPA)

Nosanov, Jeffrey: Solar System Escape Architecture for Revolutionary Science (SSEARS)

Predina, Joseph: NIST in Space: Better Remote Sensors for Better Science

Quadrelli, Marco: Orbiting Rainbows: Optical Manipulation of Aerosols and the Beginnings of Future Space Construction

Saif, Babak: Atom Interferometry for detection of Gravity Waves-a

Winglee, Robert: Sample Return Systems for Extreme Environments

Zha, GeCheng: Silent and Efficient Supersonic Bi-Directional Flying Wing

2013 NIAC project selections

In 2013 NIAC conducted a third solicitation for proposals, with projects to start in the summer of 2013. NASA selected 12 phase-I projects with a wide range of imaginative concepts, including 3-D printing of biomaterials, such as arrays of cells; using galactic rays to map the insides of asteroids; and an "eternal flight" platform that could hover in Earth's atmosphere, potentially providing better imaging, Wi-Fi, power generation, and other applications. They selected 6 phase II projects, including photonic laser thrusters, extreme sample return, and innovative spherical robots designed for planetary exploration.

Phase I selections were:

Adams, Rob: Pulsed Fission-Fusion (PuFF) Propulsion System

Bradford, John: Torpor Inducing Transfer Habitat For Human Stasis To Mars

Hemmati, Hamid: Two-Dimensional Planetary Surface Landers

Jerred, Nathan: Dual-mode Propulsion System Enabling CubeSat Exploration of the Solar System

Longman, Anthony: Growth Adapted Tensegrity Structures - A New Calculus for the Space Economy

Moore, Mark: Eternal Flight as the Solution for 'X'

Prettyman, Thomas: Deep Mapping of Small Solar System Bodies with Galactic Cosmic Ray Secondary Particle Showers

Rothschild, Lynn: Biomaterials out of thin air: in situ, on-demand printing of advanced biocomposites

Rovey, Joshua: Plasmonic Force Propulsion Revolutionizes Nano/PicoSatellite Capability

Stoica, Adrian: Transformers For Extreme Environments

2014 NIAC project selections

In 2014, NIAC conducted a fourth solicitation, and selected 12 projects for Phase-1 studies and 5 projects to continue on to phase II projects. Projects selected include a study of hibernation for astronauts and a submarine operating on Saturn's moon Titan

2014 Phase I selections were:

Atchison, Justin: Swarm Flyby Gravimetry

Boland, Eugene: Mars Ecopoiesis Test Bed

Cash, Webster: The Aragoscope: Ultra-High Resolution Optics at Low Cost

Chen, Bin: 3D Photocatalytic Air Processor for Dramatic Reduction of Life Support Mass & Complexity

Hoyt, Robert: WRANGLER: Capture and De-Spin of Asteroids and Space Debris

Matthies, Larry: Titan Aerial Daughtercraft

Miller, Timothy: Using the Hottest Particles in the Universe to Probe Icy Solar System Worlds

Nosanov, Jeffrey: PERISCOPE: PERIapsis Subsurface Cave Optical Explorer

Oleson, Steven: Titan Submarine: Exploring the Depths of Kraken

Ono, Masahiro: Comet Hitchhiker: Harvesting Kinetic Energy from Small Bodies to Enable Fast and Low-Cost Deep Space Exploration

Streetman, Brett: Exploration Architecture with Quantum Inertial Gravimetry and In Situ ChipSat Sensors

Wiegmann, Bruce: Heliopause Electrostatic Rapid Transit System (HERTS)

2015 NIAC project selections

The 2015 Phase-1 projects included a hopping vehicle to visit Triton and others, and seven phase two projects. Phase I projects selected were:

Engblom, William: Virtual Flight Demonstration of Stratospheric Dual-Aircraft Platform

Graf, John: Thirsty Walls - A new paradigm for air revitalization in life support

Hecht, Michael: A Tall Ship and a Star to Steer Her By

Lewis, John: In-Space Manufacture of Storable Propellants

Lubin, Philip: Directed Energy Propulsion for Interstellar Exploration (DEEP-IN)

Oleson, Steven: Triton Hopper: Exploring Neptune's Captured Kuiper Belt Object

Peck, Mason: Soft-Robotic Rover with Electrodynamic Power Scavenging

Plescia, Jeffrey: Seismic Exploration of Small Bodies

Paxton, Larry: CRICKET: Cryogenic Reservoir Inventory by Cost-Effective Kinetically Enhanced Technology

Sercel, Joel: APIS (Asteroid Provided In-Situ Supplies): 100MT Of Water from a Single Falcon 9

Stoica, Adrian WindBots: persistent in-situ science explorers for gas giants

Tabirian, Nelson: Thin-Film Broadband Large Area Imaging System

Ulmer, Melville: Aperture: A Precise Extremely large Reflective Telescope Using Re-configurable Elements

Wang, Joseph: CubeSat with Nanostructured Sensing Instrumentation for Planetary Exploration

Youngquist, Robert: Cryogenic Selective Surfaces

In addition, seven projects were selected for continuation into Phase II:

Atchison, Justin: Swarm Flyby Gravimetry

Chen, Bin: 3D Photocatalytic Air Processor for Dramatic Reduction of Life Support Mass and Complexity

Nosanov, Jeffrey: PERISCOPE: PERIapsis Subsurface Cave Optical Explorer

Oleson, Steven: Titan Submarine: Exploring the Depths of Kraken Mare

Paul, Michael: SCEPS in Space - Non-Radioisotope Power Systems for Sunless Solar System Exploration Missions

Stoica, Adrian: Trans-Formers for Lunar Extreme Environments: Ensuring Long-Term Operations in Regions of Darkness and Low Temperatures

Wiegmann, Bruce: Heliopause Electrostatic Rapid Transit System (HERTS)

2016 NIAC project selections

Phase I projects selected were:

Bayandor, Javid: Light Weight Multifunctional Planetary Probe for Extreme Environment Exploration and Locomotion

Bugga, Ratnakumar: Venus Interior Probe Using In-situ Power and Propulsion (VIP-INSPR)

Dunn, Jason: Reconstituting Asteroids into Mechanical Automata

Hughes, Gary: Molecular Composition Analysis of Distant Targets

Janson, Siegfried: Brane Craft

Mann, Chris: Stellar Echo Imaging of Exoplanets

Mueller, Robert: Mars Molniya Orbit Atmospheric Resource Mining

Ono, Masahiro: Journey to the Center of Icy Moons

Quadrelli, Marco: E-Glider: Active Electrostatic Flight for Airless Body Exploration

Rothschild, Lynn: Urban biomining meets printable electronics: end-to-end destination biological recycling and reprinting

Sauder, Jonathan: Automaton Rover for Extreme Environments (AREE)

Thomas, Stephanie: Fusion-Enabled Pluto Orbiter and Lander

VanWoerkom, Michael: NIMPH: Nano Icy Moons Propellant Harvester

In addition, eight projects were selected for continuation into Phase II:

Bradford, John: Advancing Torpor Inducing Transfer Habitats for Human Stasis to Mars

Engblom, William: Flight Demonstration of Novel Atmospheric Satellite Concept

Kirtley, David: Magnetoshell Aerocapture for Manned Missions and Planetary Deep Space Orbiters

Lubin, Philip: Directed Energy for Interstellar Study

Rovey, Joshua: Experimental Demonstration and System Analysis for Plasmonic Force Propulsion

Skelton, Robert: Tensegrity Approaches to In-Space Construction of a 1g Growable Habitat

Ulmer, Melville: Further Development of Aperture: A Precise Extremely Large Reflective Telescope Using Re-configurable Elements

Youngquist, Robert: Cryogenic Selective Surfaces

2017 NIAC project selections

The fifteen projects selected for Phase I were:

Adam Arkin: A Synthetic Biology Architecture to Detoxify and Enrich Mars Soil for Agriculture

John Brophy: A Breakthrough Propulsion Architecture for Interstellar Precursor Missions

John-Paul Clarke : Evacuated Airship for Mars Missions

Heidi Fearn: Mach Effects for In Space Propulsion: Interstellar Mission

Benjamin Goldman : Pluto Hop, Skip, and Jump Global

Jason Gruber: Turbolift

Kevin Kempton : Phobos L1 Operational Tether Experiment (PHLOTE)

Michael LaPointe: Gradient Field Imploding Liner Fusion Propulsion System

John Lewis : Massively Expanded NEA Accessibility via Microwave-Sintered Aerobrakes

Jay McMahon: Dismantling Rubble Pile Asteroids with AoES (Area-of-Effect Soft-bots)

Raymond Sedwick: Continuous Electrode Inertial Electrostatic Confinement Fusion

Joel Sercel: Sutter: Breakthrough Telescope Innovation for Asteroid Survey Missions to Start a Gold Rush in Space

Slava Turyshev: Direct Multipixel Imaging and Spectroscopy of an exoplanet with a Solar Gravity Lens Mission

Robert Youngquist: Solar Surfing

Nan Yu: A direct probe of dark energy interactions with a solar system laboratory

In addition, seven projects were selected for continuation into Phase II:

Ratnakumar Bugga: Venus Interior Probe Using In-situ Power and Propulsion (VIP-INSPR)

Gary Hughes: Remote Laser Evaporative Molecular Absorption Spectroscopy Sensor System

Siegfried Janson: Brane Craft Phase II

Chris Mann: Stellar Echo Imaging of Exoplanets

Jonathan Sauder: Automaton Rover for Extreme Environments (AREE)

Joel Sercel: Optical Mining of Asteroids, Moons, and Planets to Enable Sustainable Human

Exploration and Space Industrialization

Stephanie Thomas: Fusion-Enabled Pluto Orbiter and Lander

2018 NIAC project selections

The sixteen projects selected for Phase I were:

Aliakbar Aghamohammadi: Shapeshifters from Science Fiction to Science Fact: Globetrotting from Titan's Rugged Cliffs to its Deep Seafloors

David Akin: Biobot: Innovative Offloading of Astronauts for More Effective Exploration

Jeffrey Balcerski: Lofted Environmental and Atmospheric Venus Sensors (LEAVES)

Sigrid Close: Meteoroid Impact Detection for Exploration of Asteroids (MIDEA)

Christine Hartzell: On-Orbit, Collision-Free Mapping of Small Orbital Debris

Chang-kwon Kang: Marsbee - Swarm of Flapping Wing Flyers for Enhanced Mars Exploration

John Kendra: Rotary Motion Extended Array Synthesis (R-MXAS)

Chris Limbach: PROCSIMA: Diffractionless Beamed Propulsion for Breakthrough Interstellar Missions

Gareth Meirion-Griffith: SPARROW: Steam Propelled Autonomous Retrieval Robot for Ocean Worlds

Hari Nayar: BALLET: BALloon Locomotion for Extreme Terrain

Lynn Rothschild: Myco-architecture off planet: growing surface structures at destination

Dmitry Savransky: Modular Active Self-Assembling Space Telescope Swarms

Nickolas Solomey: Astrophysics and Technical Study of a Solar Neutrino Spacecraft

Grover Swartzlander: Advanced Diffractive MetaFilm Sailcraft

Jordan Wachs: Spectrally-Resolved Synthetic Imaging Interferometer

Ryan Weed: Radioisotope Positron Propulsion

In addition, nine projects were selected for continuation into Phase II:

Robert Adams: Pulsed Fission-Fusion (PuFF) Propulsion Concept

John Brophy: A Breakthrough Propulsion Architecture for Interstellar Precursor Missions

Devon Crowe: Kilometer Space Telescope (KST)

Jay McMahon: Dismantling Rubble Pile Asteroids with AoES (Area-of-Effect Soft-bots)

Steven Oleson: Triton Hopper: Exploring Neptune's Captured Kuiper Belt Object

John Slough: Spacecraft Scale Magnetospheric Protection from Galactic Cosmic Radiation

Slava Turyshev: Direct Multipixel Imaging and Spectroscopy of an Exoplanet with a Solar Gravity Lens Mission

Michael VanWoerkom: NIMPH: Nano Icy Moons Propellant Harvester

James Woodward: Mach Effect for In Space Propulsion: Interstellar Mission

2019 NIAC project selections

The twelve projects selected for Phase I were:

Javid Bayandor: BREEZE- Bioinspired Ray for Extreme Environments and Zonal Exploration

Erik Brandon: Power Beaming for Long Life Venus Surface Missions

Ana Diaz Artiles: SmartSuit: A Hybrid, Intelligent, and Highly Mobile EVA Spacesuit for Next Generation Exploration Missions

Tom Ditto: Dual Use Exoplanet Telescope (DUET)

Yu Gu: Micro-Probes Propelled and Powered by Planetary Atmospheric Electricity (MP4AE)

Troy Howe: SPEAR Probe - An Ultra Lightweight Nuclear Electric Propulsion Probe for Deep Space Exploration

Noam Izenberg: RIPS: Ripcord Innovative Power System

Geoffrey Landis: Power for Interstellar Fly-by

Joel Sercel: Lunar-Polar Propellant Mining Outpost (LPMO): Affordable Exploration and

Industrialization

John Slough: Crosscutting High Apogee Refueling Orbital Navigator (CHARON) for Active Debris Removal

George Sowers: Thermal Mining of Ices on Cold Solar System Bodies

Robert Staehle: Low-Cost SmallSats to Explore to Our Solar System's Boundaries

In addition, six projects were selected for continuation into Phase II:

Tom Ditto: The High Étendue Multiple Object Spectrographic Telescope (THE MOST)

John Kendra: Rotary-Motion-Extended Array Synthesis (R-MXAS)

Chris Limbach: Self-Guided Beamed Propulsion for Breakthrough Interstellar Missions

Nickolas Solomey: Astrophysics and Technical Lab Studies of a Solar Neutrino Spacecraft Detector

Grover Swartzlander: Diffractive Lightsails

Doug Willard: Solar Surfing

Also, two projects were selected for Phase III:

William Whittaker: Robotic Technologies Enabling the Exploration of Lunar Pits

Joel Sercel: Mini Bee Prototype to Demonstrate the Apis Mission Architecture and Optical Mining Technology

2020 NIAC project selections

The sixteen projects selected for Phase I were:

Saptarshi Bandyopadhyay: LCRT - Lunar Crater Radio Telescope on the Far-Side of the Moon

John Christian: StarNAV: An Architecture for Autonomous Spacecraft Navigation by the Relativistic Perturbation of Starlight

Artur Davoyan: Extreme Metamaterial Solar Sails for Breakthrough Space Exploration

Caroline Genzale: Fueling a Human Mission to Mars

Davide Guzzetti: Flat Fabrication of Progressively Self-Assembling Space Systems

Benjamin Hockman: Gravity Poppers: Hopping Probes for the Interior Mapping of Small Solar System Bodies

Steven Howe: Pulsed Plasma Rocket: Shielded, Fast Transits for Humans to Mars

Troy Howe: High Irradiance Peltier Operated Tungsten Exo-Reflector (HI-POWER)

Gerald Jackson: Deceleration of Interstellar Spacecraft Utilizing Antimatter

Matthew Kuhns: Instant Landing Pads for Artemis Lunar Missions

Richard Linares: Dynamic Orbital Slingshot for Rendezvous with Interstellar Objects

Philip Metzger: Aqua Factorem: Ultra Low-Energy Lunar Water Extraction

Robert Moses: Advanced Aerocapture System for Enabling Faster-Larger Planetary Science & Human Exploration Missions

Eldar Noe Dobrea: Heat Exchange-Driven Aircraft for Low Altitude and Surface Exploration of Venus

Robert Romanofsky: Magneto-Inductive Communications for Ocean Worlds

Lynn Rothschild: An Astropharmacy

In addition, six projects were selected for continuation into Phase II:

David Akin: Innovative Offloading of Astronauts for More Effective Exploration

Javid Bayandor: Lightweight Multifunctional Planetary Probe for Extreme Environment Exploration and Locomotion

Troy Howe: SPEAR Probe - An Ultra Lightweight Nuclear Electric Propulsion Probe for Deep Space Exploration

Masahiro Ono: Enceladus Vent Explorer

Joel Sercel: Lunar Polar Propellant Mining Outpost (LPMO): A Breakthrough for Lunar

Exploration & Industry

Nan Yu: Gravity Observation and Dark Energy Detection Explorer in the Solar System

Also, one project was selected for continuation into Phase III:

Slava Turyshev: Direct Multipixel Imaging and Spectroscopy of an Exoplanet with a Solar Gravitational Lens Mission

2021 NIAC project selections

The sixteen projects selected for Phase I were:

Sarbajit Banerjee: Regolith Adaptive Modification System (RAMs) to Support Early Extraterrestrial Planetary Landings (and Operations)

Sigrid Close: Exploring Uranus through SCATTER: Sustained ChipSat/CubeSat Activity Through Transmitted Electromagnetic Radiation

Amelia Greig: Ablative Arc Mining for In-Situ Resource Utilization

Zachary Manchester: Kilometer-Scale Space Structures from a Single Launch

Patrick McGarey: PEDALS: Passively Expanding Dipole Array for Lunar Sounding

Quinn Morley: Autonomous Robotic Demonstrator for Deep Drilling (ARD3)

Christopher Morrison: Extrasolar Object Interceptor and Sample Return Enabled by Compact, Ultra Power Dense Radioisotope Batteries

E. Joseph Nemanick: Atomic Planar Power for Lightweight Exploration (APPLE)

Steven Oleson: A Titan Sample Return Using In-Situ Propellants

Marco Pavone: ReachBot: Small Robot for Large Mobile Manipulation Tasks in Martian Cave Environments

Ronald Polidan: FarView – An In Situ Manufactured Lunar Far Side Radio Observatory

Ethan Schaler: FLOAT — Flexible Levitation on a Track

Ethan Schaler: SWIM — Sensing with Independent Micro-swimmers

Jane Shevtsov: Making Soil for Space Habitats by Seeding Asteroids with Fungi

Charles Taylor: Light Bender

Joshua Vander Hook: Solar System Pony Express

In addition, six projects were selected for continuation into Phase II:

Saptarshi Bandyopadhyay: LCRT - Lunar Crater Radio Telescope on the Far-Side of the Moon

Lynn Rothschild: Mycotecture Off Planet

Kerry Nock: Pluto Hop, Skip, and Jump

Artur Davoyan: Extreme Solar Sailing for Breakthrough Space Exploration

Jeffrey Balcerski: Lofted Environmental Venus Sensors (LEAVES)

Peter Gural: Sutter Ultra: Breakthrough Space Telescope for Prospecting Asteroids

Also, one project was selected for continuation into Phase III:

Nickolas Solomey: Cube-Sat Space Flight Test of a Neutrino Detector

2022 NIAC project selections

The twelve projects selected for Phase I were:

Darindra Arumugam: Cryospheric Rydberg Radar

Steven Barrett: Silent, Solid-State Propulsion for Advanced Air Mobility Vehicles

Jason Benkoski: Combined Heat Shield and Solar Thermal Propulsion System for an Oberth maneuver

Elena D'Onghia: CREW HaT: Cosmic Radiation Extended Warding using the Halbach Torus

Bonnie Dunbar: The Spacesuit Digital Thread: 4.0 Manufacture of Custom High Performance Spacesuits for the Exploration of Mars

Ivan Ermanoski: Breathing Mars Air: Stationary and Portable O2 Generation

Philip Lubin: Pi – Terminal Defense for Humanity

John Mather: Hybrid Observatory for Earth-like Exoplanets (HOEE)

Marcin Pilinski: In-situ Neutral-Optics Velocity Analyzer for Thermospheric Exploration (INOVATE)

Jonathan Sauder: Starburst: A Revolutionary Under-Constrained Adaptable Deployable Structure Architecture

Sara Seager: Venus Atmosphere and Cloud Particle Sample Return for Astrobiology

Mahmooda Sultana: SCOPE: ScienceCraft for Outer Planet Exploration

In addition, five projects were selected for continuation into Phase II:

Javid Bayandor: BREEZE- Bioinspired Ray for Extreme Environments and Zonal Exploration

Zac Manchester: Kilometer-Scale Space Structures from a Single Launch

E Nemanick: Atomic Planar Power for Lightweight Exploration (APPLE)

Marco Pavone: ReachBot: Small Robot for Large Mobile Manipulation Tasks in Martian Cave Environments

Ethan Schaler: SWIM- Sensing with Independent Micro-swimmers

Also, one project was selected for continuation into Phase III:

Amber Dubill: Diffractive Solar Sailing

2023 NIAC Project Selections

The fourteen projects selected for Phase I were:

Edward Balaban: Fluidic Telescope (FLUTE): Enabling the Next Generation of Large Space Observatories

Igor Bargatin: Photophoretic Propulsion Enabling Mesosphere Exploration

Theresa Benyo: Accessing Icy World Oceans Using Lattice Confinement Fusion Fast Fission

Zachary Cordero: Bend-Forming of Large Electrostatically Actuated Space Structures

Peter Curreri: Lunar South Pole Oxygen Pipeline

Artur Davoyan: Pellet-Beam Propulsion for Breakthrough Space Exploration

Ryan Gosse: New Class of Bimodal NTP/NEP with a Wave Rotor Topping Cycle Enabling Fast Transit to Mars

Congrui Jin: Biomineralization-Enabled Self-Growing Building Blocks for Habitat Outfitting on Mars

Mary Knapp: Great Observatory for Long Wavelengths (GO-LoW)

Quinn Morley: TitanAir: Leading-Edge Liquid Collection to Enable Cutting-Edge Science

Christopher Morrison: EmberCore Flashlight: Long Distance Lunar Characterization with Intense Passive X- and Gamma-ray Source

Heidi Newberg: Diffractive Interfero Coronagraph Exoplanet Resolver (DICER): Detecting and Characterizing All Earth-Like Exoplanets Orbiting Sun-Like Stars Within 10 pc

Stephen Polly: Radioisotope Thermoradiative Cell Power Generator

Ryan Weed: Aerogel Core Fission Fragment Rocket Engine

In addition, six projects were selected for continuation into Phase II:

Darindra Arumugam: Quantum Rydberg Radar for Surface, Topography, and Vegetation

Steven Barrett: Silent, Solid-State Propulsion for Advanced Air Mobility Vehicles

Philip Lubin: PI – Planetary Defense

Christopher Morrison: The Nyx Mission to Observe the Universe from Deep Space – Enabled by EmberCore, a High Specific Power Radioisotope Electric Propulsion System

Ronald Polidan: FarView Observatory – A Large, In-Situ Manufactured, Lunar Far Side Radio Array

Lynn Rothschild: A Flexible, Personalized, On-Demand Astropharmacy

No projects were selected for continuation into Phase III.

2024 NIAC project selections

The thirteen projects selected for Phase I were:

Matthew McQuinn: Solar System-Scale VLBI to Dramatically Improve Cosmological Distance Measurements

Kenneth Carpenter: A Lunar Long-Baseline Optical Imaging Interferometer: Artemis-enabled Stellar Imager (AeSI)

Alvaro Romero-Calvo: Magnetohydrodynamic Drive for Hydrogen and Oxygen Production in Mars Transfer

James Bickford: Thin Film Isotope Nuclear Engine Rocket (TFINER)

Ge-Cheng Zha: Mars Aerial and Ground Global Intelligent Explorer (MAGGIE)

Steven Benner: Add-on to Large-scale Water Mining Operations on Mars to Screen for Introduced and Alien Life

Lynn Rothschild: Detoxifying Mars: The Biocatalytic Elimination of Omnipresent Perchlorates

Thomas Eubanks: Swarming Proxima Centauri: Coherent Picospacecraft Swarms Over Interstellar Distances

Beijia Zhang: LIFA: Lightweight Fiber-based Antenna for Small Sat-Compatible Radiometry

Ryan Sprenger: A Revolutionary Approach to Interplanetary Space Travel: Studying Torpor in Animals for Space-health in Humans (STASH)

Geoffrey Landis: Sample Return from the Surface of Venus

Peter Cabauy: Autonomous Tritium Micropowered Sensors

Aaswath Pattabhi Raman: Electro-luminescently Cooled Zero-boil-off Propellant Depots Enabling Crewed Exploration of Mars

See also

[Advanced Concepts Team](#)

[Advanced Propulsion Physics Laboratory](#)

References

External links

[NASA Institute for Advanced Concepts site at USRA](#)

[List of NIAC studies funded 1998-2007](#)

[NASA Innovative Advanced concepts site](#)

[List of NIAC studies funded 2012-present](#)